

Data Analyst Portfolio Case Study: E-Commerce Data Cleansing

Project Overview

Dataset Profile: Thousands of records combining automated tracking codes and customer text inputs.

The Problem: The raw e-commerce export file suffered from structural errors, text corruption, and data truncation. This documentation lays out the structured, step-by-step optimization workflow executed strictly inside Excel.

Data Analyst Transformation Path

- **1. Schema Isolation (Power Query):** Intercepted the dataset initialization pipeline. Manually modified column schemas to convert 16-digit credit card integers into static text formats, entirely avoiding Excel's 15-digit scientific notation truncation limit.
- **2. Structural Unpacking (Delimiters):** Executed the text-to-columns sequence by inputting the hidden line-break carriage return code (**Ctrl + J**). This cleanly isolated multi-line text strings into independent, structured fields (Street, City, State, and Zip Code).
- **3. Quantitative Purification:** Stripped qualitative measurement text strings (e.g., 'in', 'rn') from lot entries using a nested string script: `=LEFT(Cell, 2)`. The outputs were secured permanently via Paste-as-Values formatting to enable mathematical summary calculation layers.
- **4. String Regularization:** Normalized corrupted demographic data variants across thousands of index positions using uniform Excel formulas: `=TRIM(PROPER(Target))`. This completely eliminated empty buffer spaces and standardized layout title casing.
- **5. Discrepancy Auditing:** Deployed the array logic function 'Go To Special > Blanks' to pinpoint hidden data gaps instantly. Patched entire column segments simultaneously using a synchronized **Ctrl + Enter** execution block.

Business Intelligence & KPI Dashboard

With a clean data base secured, the file was routed into interactive reporting elements. A sorted horizontal bar pivot chart was constructed to group top sales revenue by occupation. Floating metric KPI shape layers were dynamically mapped to the table's grand total fields, delivering a real-time tracking interface for executives.

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